

Interacting Multipoles of Rank 2 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}^A\Phi_{\gamma\delta\epsilon\zeta}^B=R_z^{-13}\cdot\frac{1}{315}\cdot$$

$$\begin{aligned}
& +105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\zeta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\zeta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\gamma,\delta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\epsilon)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\epsilon)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\epsilon)\cdot\delta(\gamma,\delta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\beta)\cdot\delta(\epsilon,\zeta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\epsilon)\cdot\delta(\beta,\zeta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\epsilon)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\zeta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\zeta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\delta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\epsilon)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\epsilon)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\gamma\delta\epsilon\zeta}^B
\end{aligned}$$

$$(1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}^A\Phi_{\gamma\delta\epsilon\zeta}^B=R_z^{-13}\cdot\frac{1}{315}\cdot\left\{\begin{array}{l}+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\alpha\gamma\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\alpha\gamma\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\alpha\gamma\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\alpha\gamma\gamma\zeta}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\alpha\gamma\gamma\zeta}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\alpha\gamma\gamma\zeta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\beta\beta\gamma\delta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\beta\beta\gamma\delta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\beta\beta\gamma\delta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{\beta\beta\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{\beta\beta\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\alpha\beta\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\beta}^A\cdot\Phi_{\alpha\beta\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{\beta\beta\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{\beta\beta\gamma\zeta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{\beta\beta\gamma\zeta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{\beta\beta\gamma\zeta}^B\end{array}\right. \quad (2a)$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\gamma\delta\epsilon\zeta}^B \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\gamma\delta\epsilon\epsilon}^B \\
& -1890 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\gamma\delta\delta\epsilon}^B \\
& -2835 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\gamma\gamma\delta\epsilon}^B \\
& -3780 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\
& -3780 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\gamma\delta\epsilon}^B \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\gamma\delta\epsilon}^B \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\gamma\gamma\delta\delta}^B \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\beta\gamma\delta\delta}^B \\
& +945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\beta\gamma\gamma\delta}^B \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\gamma\delta\delta}^B \\
& +945 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\gamma\gamma\delta}^B \\
& +630 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\beta\gamma\delta}^B \\
& +1260 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\beta\gamma\delta}^B \\
& -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B
\end{aligned} \right] \quad (2d)
\end{aligned}$$

Simplify R:

$$\begin{aligned}
& T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^A \Phi_{\gamma\delta\epsilon\zeta}^B = R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^A \cdot \Phi_{zz\epsilon\epsilon}^B \\
& -1890 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^A \cdot \Phi_{zz\delta\delta}^B \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^A \cdot \Phi_{zz\gamma\gamma}^B \\
& -3780 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^A \cdot \Phi_{zzz\beta}^B \\
& -3780 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^A \cdot \Phi_{zzz\alpha}^B \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zzzz}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{zz}^A \cdot \Phi_{\gamma\gamma\delta\delta}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^A \cdot \Phi_{z\beta\delta\delta}^B \\
& +945 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^A \cdot \Phi_{z\beta\gamma\gamma}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^A \cdot \Phi_{z\alpha\delta\delta}^B \\
& +945 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^A \cdot \Phi_{z\alpha\gamma\gamma}^B \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\beta\beta}^B \\
& +1260 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{zz\alpha\beta}^B \\
& -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B
\end{aligned} \right] \quad (3a)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B - 945 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\epsilon\epsilon}^B \\
& -1890 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\delta\delta}^B - 2835 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\gamma\gamma}^B \\
& -3780 \cdot R_z^6 \cdot \Theta_{z\beta}^A \cdot \Phi_{zzz\beta}^B - 3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{zzz\alpha}^B \\
& -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zzzz}^B + 315 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{\gamma\gamma\delta\delta}^B \\
& +315 \cdot R_z^6 \cdot \Theta_{z\beta}^A \cdot \Phi_{z\beta\delta\delta}^B + 945 \cdot R_z^6 \cdot \Theta_{z\beta}^A \cdot \Phi_{z\beta\gamma\gamma}^B \\
& +315 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{z\alpha\delta\delta}^B + 945 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{z\alpha\gamma\gamma}^B \\
& +630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\beta\beta}^B + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{zz\alpha\beta}^B \\
& -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B
\end{aligned} \right] \quad (3b)
\end{aligned}$$

$$\xrightarrow{\text{reduced indices}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{array}{l} +10395 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B - 945 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\alpha\alpha}^B \\ -1890 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\alpha\alpha}^B - 2835 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\alpha\alpha}^B \\ -3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{zzz\alpha}^B - 3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{zzz\alpha}^B \\ -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zzzz}^B + 315 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{\alpha\alpha\beta\beta}^B \\ +315 \cdot R_z^6 \cdot \Theta_{z\beta}^A \cdot \Phi_{z\alpha\alpha\beta}^B + 945 \cdot R_z^6 \cdot \Theta_{z\beta}^A \cdot \Phi_{z\alpha\alpha\beta}^B \\ +315 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{z\alpha\beta\beta}^B + 945 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{z\alpha\beta\beta}^B \\ +630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\beta\beta}^B + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{zz\alpha\beta}^B \\ -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \end{array} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned} &+10395 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B - 5670 \cdot R_z^6 \cdot \Theta_z^A \cdot \Phi_{zz\alpha\alpha}^B \\ &- 7560 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{zzz\alpha}^B - 945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zzzz}^B \\ &+ 315 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{\alpha\alpha\beta\beta}^B + 1260 \cdot R_z^6 \cdot \Theta_{z\beta}^A \cdot \Phi_{z\alpha\alpha\beta}^B \\ &+ 1260 \cdot R_z^6 \cdot \Theta_{z\alpha}^A \cdot \Phi_{z\alpha\beta\beta}^B + 630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\beta\beta}^B \\ &+ 1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{zz\alpha\beta}^B - 45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ &- 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^A \cdot \Phi_{\alpha\beta\gamma\gamma}^B \end{aligned} \right] \quad (3d)$$

Simplify Quadrupoles:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}^A\Phi_{\gamma\delta\epsilon\zeta}^B=R_z^{-13}\cdot\frac{1}{315}\cdot\left[\begin{aligned}&+10395\cdot R_z^6\cdot\Theta_{zz}^A\cdot\Phi_{zzzz}^B-5670\cdot R_z^6\cdot\Theta_{zz}^A\cdot\Phi_{zz\alpha\alpha}^B\\&-7560\cdot R_z^6\cdot\Theta_{zz}^A\cdot\Phi_{zzzz}^B-945\cdot R_z^6\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{zzzz}^B\\&+315\cdot R_z^6\cdot\Theta_{zz}^A\cdot\Phi_{\alpha\alpha\beta\beta}^B+1260\cdot R_z^6\cdot\Theta_{zz}^A\cdot\Phi_{zz\alpha\alpha}^B\\&+1260\cdot R_z^6\cdot\Theta_{zz}^A\cdot\Phi_{zz\beta\beta}^B+630\cdot R_z^6\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{zz\beta\beta}^B\\&+1260\cdot R_z^6\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{zz\alpha\alpha}^B-45\cdot R_z^6\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{\beta\beta\gamma\gamma}^B\\&-180\cdot R_z^6\cdot\Theta_{\alpha\alpha}^A\cdot\Phi_{\alpha\alpha\gamma\gamma}^B\end{aligned}\right]\quad(4a)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned} &+10395 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B - 5670 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\alpha\alpha}^B \\ &- 7560 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B - 945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zzzz}^B \\ &+ 315 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{\alpha\alpha\beta\beta}^B + 1260 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\alpha\alpha}^B \\ &+ 1260 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\alpha\alpha}^B + 630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\beta\beta}^B \\ &+ 1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\alpha\alpha}^B - 45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B \\ &- 180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\alpha\alpha\beta\beta}^B \end{aligned} \right] \quad (4b)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned} &+2835 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B - 3150 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zz\alpha\alpha}^B \\ &-945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zzzz}^B + 315 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{\alpha\alpha\beta\beta}^B \\ &+630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\beta\beta}^B + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\alpha\alpha}^B \\ &-45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\beta\beta\gamma\gamma}^B - 180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{\alpha\alpha\beta\beta}^B \end{aligned} \right] \quad (4c)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}^A\Phi_{\gamma\delta\epsilon\zeta}^B = R_z^{-13} \cdot \frac{1}{315} \cdot [+2835 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^A \cdot \Phi_{zz\alpha\alpha}^B] \quad (5)$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}^A\Phi_{\gamma\delta\epsilon\zeta}^B = R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned} &+2835 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B + 1260 \cdot R_z^6 \cdot \Theta_{xx}^A \cdot \Phi_{xxzz}^B \\ &+ 1260 \cdot R_z^6 \cdot \Theta_{yy}^A \cdot \Phi_{yyzz}^B + 1260 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B \end{aligned} \right] \quad (6a)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[+4095 \cdot R_z^6 \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B + 1260 \cdot R_z^6 \cdot \Theta_{xx}^A \cdot \Phi_{xxzz}^B \right. \\ \left. + 1260 \cdot R_z^6 \cdot \Theta_{yy}^A \cdot \Phi_{yuzz}^B \right] \quad (6b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}^A\Phi_{\gamma\delta\epsilon\zeta}^B = \left[\begin{array}{c} +13 \cdot R_z^{-7} \cdot \Theta_{zz}^A \cdot \Phi_{zzzz}^B + 4 \cdot R_z^{-7} \cdot \Theta_{xx}^A \cdot \Phi_{xxzz}^B \\ +4 \cdot R_z^{-7} \cdot \Theta_{yy}^A \cdot \Phi_{yyzz}^B \end{array} \right] \quad (7)$$